# A Tour of tekx

*Mathematics that means something*

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**Abstract**

This document is written in tekx. It shows how prose, structure and mathematics read naturally in source form, and how one source yields typeset pages, MathML, spoken mathematics and clean copy text.

## 1 Prose and mathematics

Let us begin with the golden ratio, $\varphi = \left(1 + \sqrt{5}\right)/2$. It satisfies

$$\varphi^2 = \varphi + 1 \tag{1}$$

which a screen reader speaks as "phi squared equals phi plus 1", and which copies out of the PDF as `φ² = φ + 1`. Paste that back into a tekx file and you get the same formula.[1] Long words such as incomprehensibilities are hyphenated and still copy as one word; see [1] and `https://example.org`.

**Theorem 1** (Pythagoras)**.** *For a right triangle with legs a, b and hypotenuse c,*

$$a^2 + b^2 = c^2$$

*Proof.* Arrange four copies of the triangle inside a square of side $a + b$. ∎

**Principle 2** (Least surprise)**.** *A formula should read as it is spoken.*

> **Note.** Equation (1) determines $\varphi$ up to sign, and Theorem 1 is older than Pythagoras.

## 2 Structure

Figures, like Figure 1, are numbered and referenced like everything else.

- Lists, with $x \in \mathbb{R}$ inside.
- Items long enough to wrap across more than one line keep their hanging indentation.
  1. Nested and numbered.
  2. Like this.

[1] Copy and paste is checked with poppler, which honours ActualText, and with pdf.js, which does not.

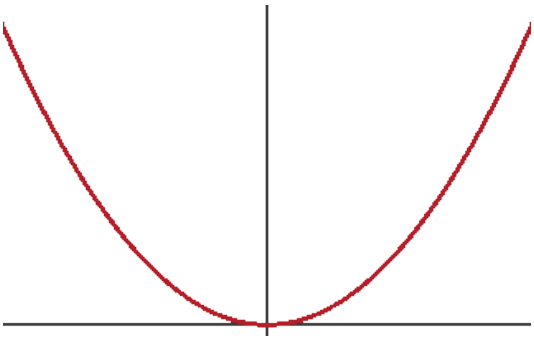


**Figure 1.** The graph of $y = x^2$. A figure must say what it shows: tekx refuses one without alt text.

| Symbol | Name | Meaning |
|---|---|---|
| $\mathbb{N}$ | NN | natural numbers |
| $\mathbb{R}$ | RR | real numbers |

```
tekx tour.tekx      # writes tour.pdf, silently
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$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x & \text{otherwise} \end{cases}$$

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ &= b^2 + 2ab + a^2 \end{aligned}$$

## References

[1] N. N. Vorobiev, *Fibonacci Numbers*, Birkhäuser, 2002.